Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

March / April 2018

EC203.01 / EC 203 Solid State Electronics

Date: 02.04.2018, Monday **Time: 1:30 p.m. to 4.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Use of scientific calculator is allowed.

SECTION-I

Q-1	**Answer Following Questions**	**[05]**
(a)	Draw Half Wave Rectifier.	[02]
(b)	Describe Diode Biasing : Forward and Reverse	[03]
Q-2	**Attempt Any Three**	**[15]**
(a)	Draw and explain circuit of fixed bias (base resistor) method for transistor	[05]
(b)	Explain the working of NPN transistor with necessary circuit diagram.	[05]
(c)	Draw and explain circuit of voltage divider bias method for transistor	[05]
(d)	How transistor works as an amplifier?	[05]
Q-3	**Attempt Any Three**	**[15]**
(a)	Explain dc load line with Q-point.	[05]
(b)	Draw Common Emitter output characteristics.	[05]
(c)	Give the difference between FET and BJT	[05]
(d)	Explain FET construction, operation and static characteristics	[05]

SECTION – II

Q-4	**Attempt Following Questions**	**[05]**
(a)	Difference between JFET and MOSFET	[02]
(b)	Explain Pinch off Voltage and Gate to Source Voltage.	[03]
Q-5	**Attempt Any Three**	**[15]**
(a)	Draw and explain circuit diagram of direct coupled multi stage amplifier	[05]

(b)	Draw and explain circuit diagram of RC coupled multi stage amplifier.	[05]
(c)	What is stabilization in transistor? Describe the need of stabilization in detail	[05]
(d)	What is stability factor in transistor? Derive mathematical expression of stability factor in transistor.	[05]
(e)	Draw input and output characteristics of common base (CB) transistor connection.	[05]
Q-6	**Attempt Any Three**	**[15]**
(a)	Explain the construction and working of metal oxide semiconductor field effect transistor (MOSFET).	[05]
(b)	Define: Frequency Response, Decibel Gain and Bandwidth Gain.	[05]
(c)	Describe Frequency Distortion.	[05]
(d)	Describe Shunt capacitor filter and Choke Input Filter	[05]
(e)	Describe Diode in open circuited condition with junction theory and barrier potential concept	[05]

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